

Artificial Intelligence Term work

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Problem Definition: -

Project Title: RoadGuard AI – Smart Road Damage Detection and Assessment System

Road damage such as potholes and cracks can lead to road accidents, vehicle damage, traffic problems, and increased maintenance costs. Traditional road inspection methods rely heavily on manual monitoring, which can be time-consuming, costly, and difficult to perform frequently across large road networks.

This project aims to develop a Road Damage Detection and Assessment System using Image Processing and Deep Learning techniques that automatically identifies road damage from images captured using a camera or mobile device. The system will use image preprocessing techniques and a Convolutional Neural Network (CNN) to classify different types of road damage such as potholes, longitudinal cracks, transverse cracks, and alligator cracks.

The system will analyze the detected damage, assign different severity levels, calculate a road condition score, and provide maintenance priority and recommendations. The proposed solution can help road authorities identify damaged roads efficiently, improve road maintenance planning, reduce inspection time, and support timely action to improve road safety.